# Predictive Analysis of Landslide Susceptibility under Hydrological Aspects of Climate Change in Kegalle District, Sri Lanka

Lumin Gunasinghe
*Department of Civil Engineering*
*University of Moratuwa*
Colombo, Sri Lanka
180210t@uom.lk

Luminda Gunawardhana
*Department of Civil Engineering*
*University of Moratuwa*
Colombo, Sri Lanka
lumindang@uom.lk

Lalith Rajapakse
*Department of Civil Engineering*
*University of Moratuwa*
Colombo, Sri Lanka
lalith@uom.lk

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***Abstract*—Climate change is widely recognized as a global phenomenon that has far-reaching consequences, including an increase in the frequency and severity of landslides. In response to this global concern, this study narrows its focus to predict landslide susceptibility in the Kegalle District in Sri Lanka, specifically, emphasizing the hydrological aspects influenced by climate change within this region. Potential changes in future rainfall regimes were projected using the HadGEM3-GC31-LL model from the Coupled Model Intercomparison Project Phase 6 (CMIP6). Coarse-resolution rainfall from this model was statistically downscaled to the meteorological station level using the Long Ashton Research Station Weather Generator, known as LARS-WG. In addition to the rainfall influence, the effects of watershed slope, elevation, soil type, land use and land cover type, and the distance from the river were also considered in the Analytical Hierarchy Process (AHP). By analyzing historical landslide occurrences, results demonstrated a concordance of 50%-70% between observed landslide occurrences and rainfall effects. The incorporation of other landslide-triggering factors improved the accuracy by up to 86%. Subsequently, future landslide maps were generated. The output will aid the decision-makers in prioritizing mitigation efforts and disaster-resilient urban planning.**

***Keywords*—*extreme rainfall, geohazards, downscaling, AHP analysis, GIS***

## I. INTRODUCTION

Landslide occurrences have increased globally causing even more economic losses and fatalities in the recent past. Landslides accounted for 1.3% of natural hazard fatalities and 4.9% of natural disaster events between 1990 and 2015 worldwide [1]. More than 18,000 deaths occurred due to landslides from 1998-2017, affecting 4.8 million people globally. Asia is the most affected region with 54% of landslides occurring there. Landslides have caused around 8 billion US dollars in economic losses worldwide [2]. It is widely acknowledged that landslides pose a significant threat to people's lives and negatively impact a country's economy.

Landslides can be triggered by various factors such as steep slopes, geology, land use, seismic and human activities, and extreme precipitation [3,4]. Rainfall, in particular, is an important factor that can increase landslide susceptibility under changing climate conditions. The Intergovernmental Panel on Climate Change (IPCC) predicts that global warming will cause more frequent and intense rainfall and drought events, which can further increase the likelihood of landslides [5]. Therefore, it is essential to consider these changes when assessing landslide susceptibility.

Global circulation models (GCMs) are used to simulate Earth's climate system thereby projecting potential changes in the future [6]. However, the application of GCM results directly at a local level is problematic due to resolution mismatches. To make GCM outputs usable on local scales, downscaling approaches such as dynamic and statistical downscaling are used [7]. The Long Ashton Research Station Weather Generator (LARS-WG) is one of these statistical downscaling software commonly used for climate change studies [8].

United Nations Office for Disaster Risk Reduction (UNDRR) defines a disaster as an event with a serious disruption to a community or society that results in widespread losses and exceeds the affected community's ability to cope. Disaster management aims to limit losses, provide aid, and facilitate rapid recovery through a cycle of activities: mitigation, preparation, response, and recovery. Mitigation involves analyzing hazards and reducing their potential impact, making it a crucial proactive measure in disaster management, especially for preventing or minimizing the impact of landslides.

Landslide susceptibility refers to the likelihood of a landslide occurrence in a specific area based on topography and hydrological conditions, and it can be used to create landslide susceptibility maps [9]. These maps can be developed using statistical and GIS methods, such as landslide susceptibility analysis, and used as disaster prevention and mitigation tools. They are particularly useful in regional land use planning, site selection for construction activities, and decision-making to prevent disasters [10]. With increasing urbanization in Sri Lanka, it is important to identify landslide-prone locations to implement control measures and prevent disasters.

There are two methods for mapping landslide susceptibility, qualitative and quantitative. The qualitative method relies on expert knowledge and is simpler than the quantitative method, which involves ranking and weighting factors using methods like the Analytic Hierarchy Process (AHP). The AHP method is commonly used in landslide susceptibility analysis to create a final map by assembling layers of various factors weighted according to their significance [11].

This study utilizes the AHP method to develop a comprehensive landslide susceptibility map and the primary goal is to identify suitable land use planning and management strategies, as well as to make accurate forecasts, ultimately leading to a reduction in potential losses caused by landslides. It is essential to emphasize the urgency of this undertaking due to the increasing frequency

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of landslides triggered by hydro-meteorological events in Sri Lanka, particularly in the central highlands. These events have inflicted substantial damage and destruction, negatively impacting the country's Gross Domestic Product (GDP) and the well-being of those residing in affected areas. Consequently, the mitigation of landslide risk has emerged as a critical and immediate imperative for the Sri Lankan administration [12].

Sri Lanka has about 20% of its land mass covered by mountainous and hilly terrain with high-grade metamorphic rocks rich in feldspar and flaky minerals. Despite being in a tropical environment, Sri Lanka experiences significant rates of weathering and soil formation, resulting in thick, unstable soil profiles that trigger landslides during heavy rainfall. The central highland region, consisting of 12 districts, is particularly prone to landslides, with frequent occurrences causing significant loss of life, including nearly 1,000 human lives between 2003-2016 [12].

For this research, the Kegalle district was selected as the study area located in the central highland region of Sri Lanka. Landslides were less common in this area a few decades ago, but they have become more frequent due to changes in climate and human activities [13,14]. In 2018, research investigations found that 13% of the entire area was highly susceptible to landslides, and 37% of the total land area was prone to slope failures [14]. In 2016, a disastrous situation occurred in the Kegalle district resulting in the displacement of 168 families, and nearly 1,930 families were evacuated as a precautionary measure due to high landslide risk. Around 2,000 families were planned to be resettled as the government decided to provide permanent shelters to the displaced families [15].

The frequent occurrence of landslides in Sri Lanka's central highlands has led to the need for implementing appropriate adaptive measures. Studies have shown that the annual total rainfall in many regions of Sri Lanka has increased in recent decades, which may be a contributing factor to the increase in landslides [16]. However, with changing climate conditions, the exclusive use of historical data may not be applicable in predicting future situations. Therefore, new predictive measures should be adapted with the use of new climate projections to reduce future damages.

### *A. Objectives*

#### *1) Main Objective*

The main objective of this research is to estimate landslide susceptibility variations attributed to climate change in the Kegalle district.

#### *2) Specific Objectives*

- To assess the impacts of climate change on local rainfall patterns and variations
- To produce a landslide susceptibility map using the AHP method for the focused area considering all possible influential factors
- To integrate extreme rainfall variations in the future with the AHP analysis to quantify landslide susceptibility variations under changing climate conditions

## II. Study Area

The Kegalle district is located in the wet and intermediate zones of the country. A catchment area of 1,523 $km^2$ in the Kegalle district was selected as the study area (Fig. 1). The area experiences heavy rainfall during the rainy season and bright sunshine during the dry season. The average annual rainfall is 2,500 mm to 3,000 mm and the average annual temperature varies from 25°C to 28°C.

Home gardens cover 35% of the land in the district, and rubber is the main crop with 31% of the land cover. Other main crops include tea, coconut, and paddy. The district is geologically situated in the western plains of the central highland and most soils have developed from residual materials derived by weathering of the bedrock or in the weathering materials that have been transformed for a short distance. Red-yellow podzolic can be categorized as the dominant soil group in the district [14].

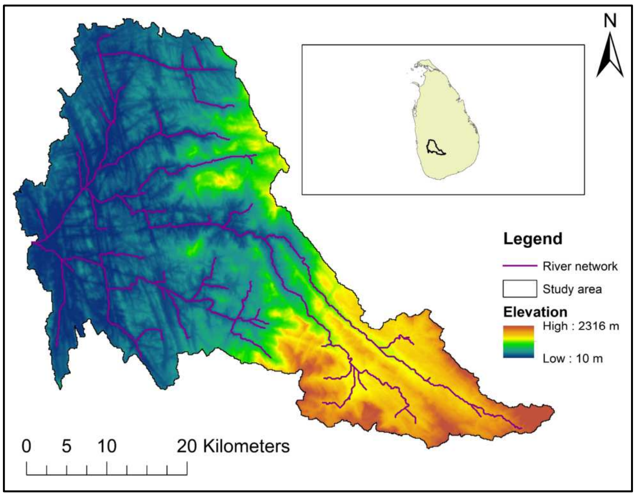

Fig. 1. Study area

## III. Methodology

### *A. Rainfall Downscaling*

Potential changes in future rainfall regimes were projected using the HadGEM3-GC31-LL model from the Coupled Model Intercomparison Project Phase 6 (CMIP6) [5,6]. Coarse-resolution rainfall from this model was statistically downscaled to the meteorological station level using the Long Ashton Research Station Weather Generator, known as LARS-WG [8]. The baseline period selected for analysis is 1975-2015, while two Shared Socioeconomic Pathways: SSP2-4.5 and SSP5-8.5 were considered for future projections. Considering the long-term data required for the statistical downscaling, the Ratnapura rain gauge station was selected ground controlling point for the analysis.

### *B. Thematic Data Layers*

#### *1) Rainfall*

The missing data percentage in Ratnapura was 0.20% during the baseline period. The Multivariate Imputation by Chained Equations (MICE) method was then used to fill in the missing values using the data from Annfield, Laxapana, and Nuwara Eliya stations [17]. Extreme Value Type I (Gumble distribution) was used to develop a rainfall intensity-frequency relationship and compared with landslide data received from the National Building Research Organization (NBRO) (Fig. 2). A threshold of 73 mm per day was identified as the minimum rainfall amount required to trigger past landslides in the area by analyzing the graph (Fig. 2.) The number of days above this threshold was then used as a landslide-triggering factor.

The downscaled daily rainfalls at Ratnapura station were distributed to the study area using a grided rainfall dataset known as Climate Hazards group InfraRed Precipitation with Station (CHIRPS) with a grid resolution of 5 km x 5 km. The catchment area covers 89 grid cells (Fig. 3). A Python code was developed to produce bias-correction factors between observed rainfalls and the corresponding CHIRPS values for each month and cell. Assuming these factors will remain unchanged (i.e. law of stationarity), a time series of future projections for each grid cell was created. The distribution of the average number of days per year with daily rainfall of more than 73 mm was mapped (Fig. 4).

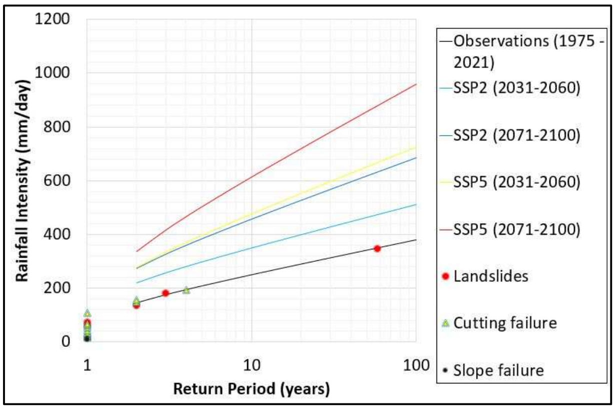


Fig. 2. Rainfall intensity vs return period graph

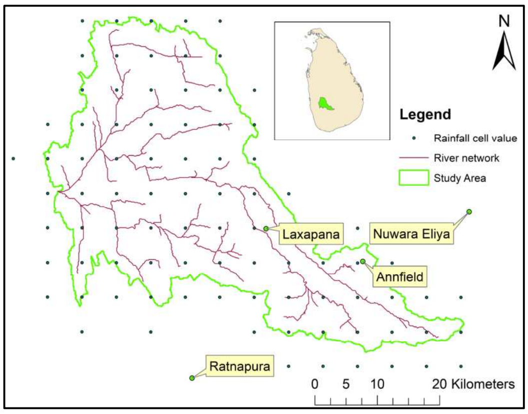


Fig. 3. The coverage of CHIRPS data

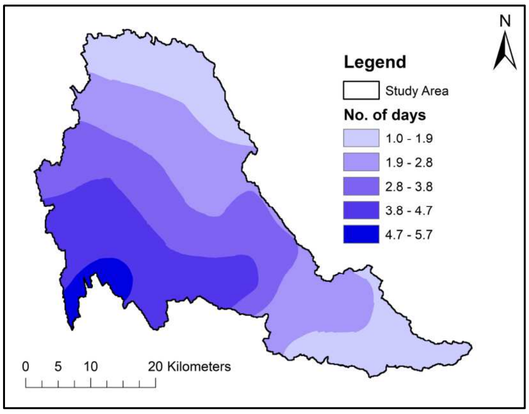


Fig. 4. Number of days per year exceeding the threshold rainfall value during the baseline period (1975-2015)

*2) Elevation*

The ground elevation was assigned using ArcGIS 10.8 from the freely available 30-meter Shuttle Radar Topography Mission (SRTM) data downloaded from the Earth Explorer (Fig. 5). The catchment area exhibits a wide range of elevations, spanning from 10 – 2,316 m above sea level.

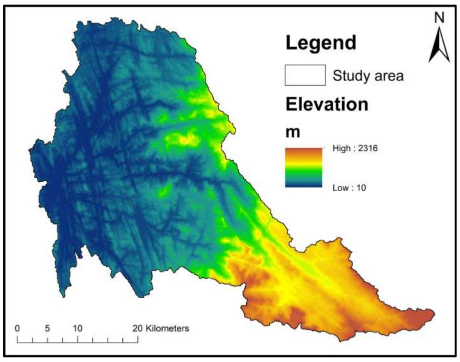


Fig. 5. Elevation map

*3) Slope*

The steepness of a slope is a significant factor that can cause or initiate landslides. Due to the increased forces caused by gravity, steeper slopes are more prone to landslides [18]. The slope tool in ArcGIS was used to derive the slope angle (Fig. 6).

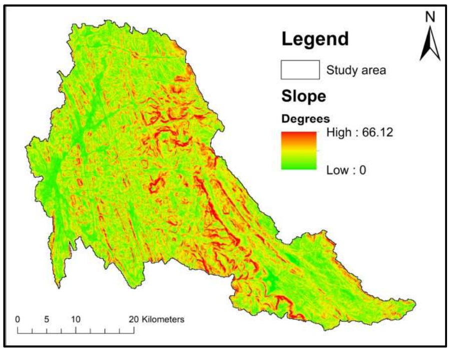


Fig.6. Slope map

*4) Soil type*

The characteristics of the topsoil are another triggering factor that affects landslides. Soil properties like permeability, hydraulic conductivity, and cohesion vary depending on the type of soil. To obtain the soil map for the study area, the geo-referenced thematic soil map provided by the National Spatial Data Infrastructure Sri Lanka (NSDI) was utilized. Four main soil types were identified (Fig. 7).

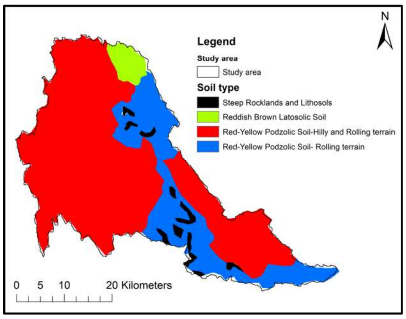


Fig. 7. Soil map

*5) Land Use and Land Cover (LULC)*

Land cover plays a significant role in influencing the stability of slopes and regulating water content within the slopes. Additionally, the presence of vegetation, through its root systems, contributes to reinforcing the slope. The impact of land cover on landslide susceptibility is crucial, as it aids in absorbing soil moisture and reducing the potential for landslides [19]. Consequently, land cover is regarded as a fundamental factor in the preparation of landslide susceptibility maps. For this research, the LULC map was acquired from Sentinel-2 imagery with a resolution of 10 meters, provided by Esri. The downloaded LULC map was then categorized into five classes (Fig. 8).

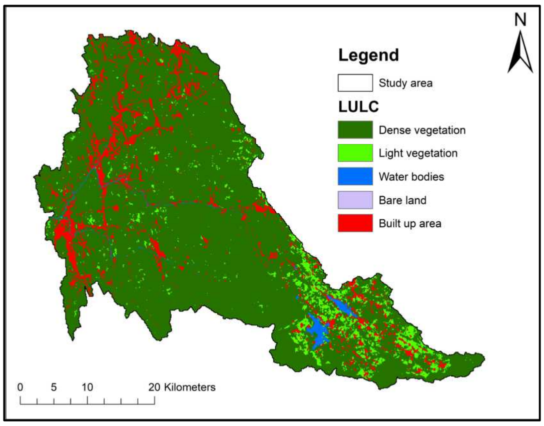


Fig. 8. LULC Map

*6) Distance from the river*

The relationship between landslides and proximity to the river is inversely related. As one gets closer to the river, the likelihood of slope instability increases. The river network data was obtained from the SRTM file and processed using the Arc Hydro Tools in ArcMap. By employing the Multiple-Ring Buffer tool, the distance from the river was categorized into five buffer zones.

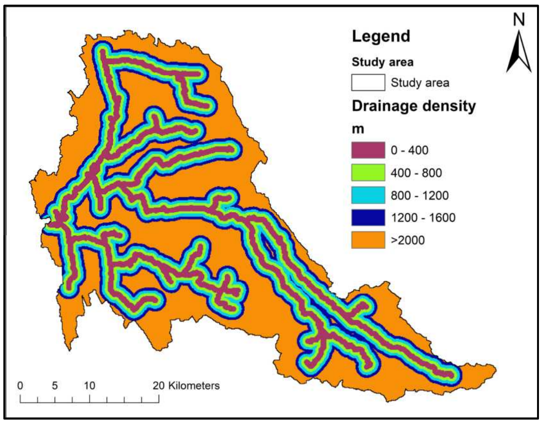


Fig. 9. Distance from the river map

## IV. Results and Discussion

*A. Analysis of the Rainfall Data*

TABLE I presents the rainfall intensity values (in mm/day) for a 100-year return period for different scenarios. The observed rainfall intensity, representing the baseline condition, is 378.8 mm/day. The subsequent row indicates the percentage increase in rainfall intensity compared to the observed value.

For the SSP2 scenario, the rainfall intensity shows a 35.0% increase for the period of 2031-2060 and a significant 80.5% increase for the period of 2071-2100. Similarly, under the SSP5 scenario, the rainfall intensity exhibits a substantial 91.1% increase for the period of 2031-2060 and a remarkable 152.6% increase for the period of 2071-2100. These findings highlight a clear upward trend in rainfall intensity for both SSP2 and SSP5 scenarios, indicating a potential future increase in extreme precipitation events.

TABLE I. Comparison Of The Hundred Year Return Period Rainfall Values For Different Scenarios

| **Scenarios** | **Observations** | **SSP2 (2031 - 2060)** | **SSP2 (2071 - 2100)** | **SSP5 (2031 - 2060)** | **SSP5 (2071 - 2100)** |
|---|---|---|---|---|---|
| Rainfall intensity (mm/day) | 378.8 | 511.2 | 683.7 | 723.9 | 957.0 |
| Percentage increase (%) | – | 35.0 | 80.5 | 91.1 | 152.6 |

Fig. 10 depicts the distribution of the average number of days per year with daily rainfall of more than 73 mm during the 2016-2021 period and the locations of landslides, slope failure, and cutting failure that occurred during the same period. This 73 mm threshold was identified as the minimum rainfall amount required to trigger past landslides in the area.

The number of days above this threshold was then used as an indicator with the observed events to define the range of days/year required for triggering landslides, slope failures, and cutting failures (Table II).

TABLE II. Accuracy Of The Results Based On Past Events

| **Category** | **Number of Events** | **Range of Days** | **Accuracy** |
|---|---|---|---|
| Landslide | 10 | > 4.2 | 70% |
| Cutting Failure | 15 | 2.0-4.2 | 60% |
| Slope failure | 4 | < 2.0 | 50% |

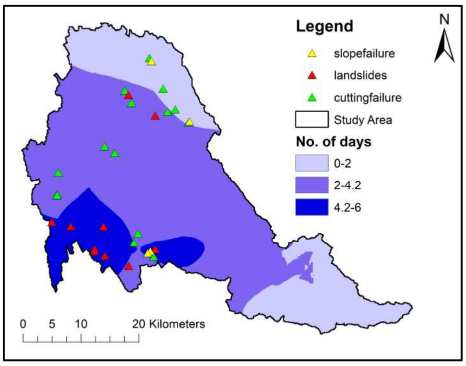


Fig. 10. Distribution of the number of days exceeding the threshold

*B. AHP Analysis*

The weights of each of the six factors (Table III) were calculated using the pair-wise matrix and the consistency ratio (CR) came out as 0.035. The (CR) was calculated to assess the reliability and consistency of the pairwise comparisons. The obtained CR value is significantly below the acceptable threshold of 0.1. This indicates a high level of consistency in the judgments made during the analysis. A low CR value is crucial as it ensures that the weights assigned to various criteria and factors accurately reflect their relative importance. Consequently, it enhances the credibility and reliability of the analysis results. The obtained CR value of 0.035 reinforces the validity of the pairwise comparisons made in the AHP analysis and instills confidence in the accuracy of the final landslide susceptibility assessment. Finally, after integrating all the landslide triggering factors, the final landslide susceptibility map was produced. It showed an accuracy of 86% with 25 out of 29 events falling into areas identified as high risk or very high risk (Fig. 11)

TABLE III. LANDSLIDE TRIGGERING FACTORS AND ASSIGNED WEIGHTS

| Triggering Factors | Class | Risk | Weightage |
|---|---|---|---|
| Number of days per year exceeding the threshold rainfall (days) | 0-1.4 | Very Low | 24 |
| | 1.4 – 2.3 | Low | |
| | 2.3 – 3.3 | Moderate | |
| | 3.3 – 4.2 | High | |
| | >4.2 | Very High | |
| Slope (degrees) | 0 – 13.22 | Very Low | 11 |
| | 13.22 – 26.45 | Low | |
| | 26.45 – 39.67 | Moderate | |
| | 39.67 – 52.90 | High | |
| | 52.90 – 66.12 | Very High | |
| Elevation (m) | 10 – 471 | Very Low | 5 |
| | 471 – 932 | Low | |
| | 932 – 1,394 | Moderate | |
| | 1,394 – 1,855 | High | |
| | 1,855 – 2,316 | Very High | |
| Distance from the River (m) | 0 – 400 | Very High | 35 |
| | 400- 800 | High | |
| | 800 – 1,200 | Moderate | |
| | 1,200 – 1,600 | Low | |
| | >2,000 | Very Low | |
| Soil Type | Steep Rocklands and Lithosols | Low | 18 |
| | Reddish Brown Latosolic soil | Moderate | |
| | Red-Yellow Podzolic soil- Rolling terrain | High | |
| | Red-Yellow Podzolic soil- Hilly and Rolling terrain | Very High | |
| LULC | Dense Vegetation | Very Low | 7 |
| | Light Vegetation | Low | |
| | Water Bodies | Moderate | |
| | Bare Land | High | |
| | Built up Area | Very High | |

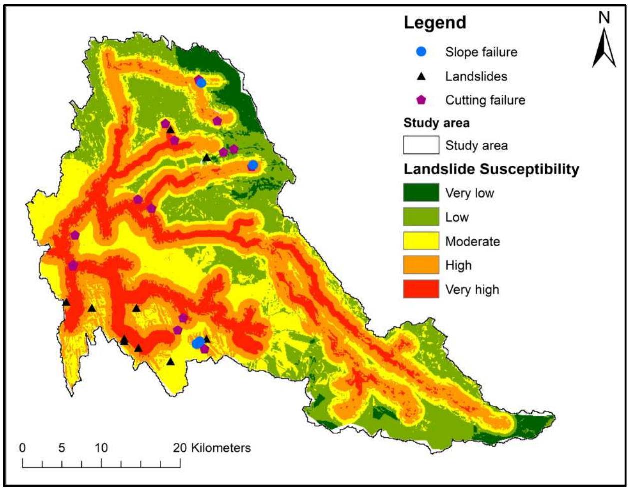


Fig. 11. Final landslide susceptibility map

TABLE IV illustrates the percentage distribution of each class across the entire catchment. According to the results, the majority of the catchment area falls into the high susceptibility category, accounting for 31.35% of the total area. The moderate susceptibility class encompasses 24.92% of the catchment, followed by the low susceptibility class at 23.63%. The very low susceptibility class covers a relatively small portion of the catchment, representing 4.51% of the area. Lastly, the very high susceptibility class comprises 15.59% of the total catchment area. This comparison provides valuable insights into the distribution of landslide susceptibility within the catchment, highlighting areas of higher vulnerability.

TABLE IV. COMPARISON OF THE SUSCEPTIBILITY CLASSES

| Susceptibility Class | Very Low | Low | Moderate | High | Very High |
|---|---|---|---|---|---|
| Area coverage of each class (%) | 4.51 | 23.63 | 24.92 | 31.35 | 15.59 |

Fig. 12 shows the potential changes of the landslide susceptibility as estimated from the projected rainfall by SSP2-4.5 during the 2031-2060 period. Results depict that the distribution of High and Very high susceptibility regions expand throughout the catchment area in the future compared to the observational period. Similar effects were observed for both SSP scenarios for the period of 2031-2060 (Table V).

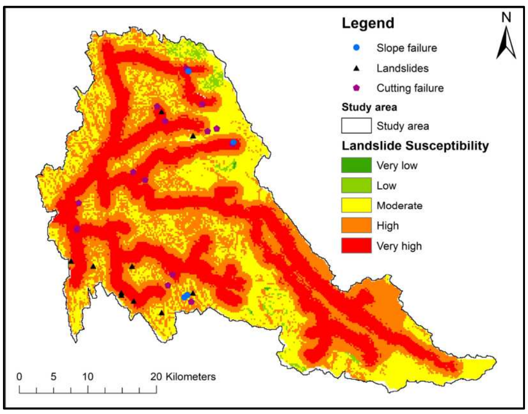


Fig. 12. Predictive landslide susceptibility map for SSP2-4.5 (2031-2060)

TABLE V. COMPARISON OF THE LANDSLIDE SUSCEPTIBILITY

| Scenarios | Observations | SSP2 (2031-2060) | SSP5 (2031-2060) |
|---|---|---|---|
| Area coverage of the extreme regions (High and Very high) (%) | 46.95 | 71.57 | 72.97 |

TABLE VI illustrates the percentage distribution of each class across the entire catchment of the future predictive map. The very low susceptibility class encompasses the smallest percentage, accounting for only 0.04% of the total area. The low susceptibility class covers a slightly larger portion, representing 1.16% of the study area. The moderate susceptibility class constitutes a significant portion, encompassing 27.23% of the total area. The high susceptibility class comprises the largest percentage, with 39.47% of the study area falling into this category. Finally, the very high susceptibility class represents 32.10% of the total area. These findings indicate that a considerable portion of the study area is highly susceptible to landslides in future scenarios, emphasizing the importance of implementing appropriate mitigation measures and land-use planning strategies to minimize potential risks and ensure the safety of the affected areas.

TABLE VI. COMPARISON OF THE SUSCEPTIBILITY CLASSES OF THE PREDICTIVE MAP

| Susceptibility Class | Very Low | Low | Moderate | High | Very High |
|---|---|---|---|---|---|
| Area coverage of each class (%) | 0.04 | 1.16 | 27.23 | 39.47 | 32.10 |

## V. CONCLUSION

The findings of this study indicate that the rainfall effect has a significant influence on landslide incidents, with a match rate of 50% to 70% with past landslide events. By incorporating the other five (5) triggering factors, the accuracy of the susceptibility map improves to 86%. The analysis of three different scenarios and their respective observations provides valuable insights into the expansion of susceptible areas (Table V). The results signify a substantial proportion of susceptible areas and highlight the need for proactive measures to address the associated risks. However, the future scenarios demonstrate a more alarming trend. The outcomes of this study, therefore, provide valuable insights for relevant authorities and decision-makers, enabling them to incorporate hydrological factors alongside other landslide triggers in order to identify areas prone to future landslides. This knowledge can guide the development of infrastructure and appropriate measures in these identified susceptible areas.

Also, the results of this study suggest several prospective avenues for future research in the fields of landslide susceptibility assessment and the influence of climate change. Important focal points include the need to improve climate models in regions susceptible to landslides, with an aim to enhance the precision of predictions and reduce uncertainties within the models. Moreover, it is recommended to strengthen the reliability of predictive models by validating them using additional historical landslide occurrence data.

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